

ZHU JIN

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EDUCATION

- School of Physics, Georgia Institute of Engineering, Atlanta, GA.** Aug 2025 - present
Ph.D. Student of Quantitative Biosciences (Physics)
Advisor: Dr. Peter Yunker
- UM-SJTU Joint Institute, Shanghai Jiao Tong University, Shanghai, China.** Sep 2023 - Jul 2025
Master of Science in Mechanical Engineering
Thesis title: “Bacterial Locomotion in Swimming and Swarming Motilities” - Supervised by Prof. Zijie Qu
Committee members: Dr. Zijie Qu, Dr. Wendong Wang, Dr. Yunlong Guo
- UM-SJTU Joint Institute, Shanghai Jiao Tong University, Shanghai, China.** Sep 2019 - Aug 2023
Bachelor of Engineering in Mechanical Engineering

PUBLICATION

- Duan, S., Wang, Z., Xiong, K., **Zhu, J.**, Gu, P., Chen, W., Xin, H., and Qu, Z. **From shape to fate: making bacterial swarming expansion predictable.** *arXiv preprint.* [arXiv:2602.01056](https://arxiv.org/abs/2602.01056). (Under Review)
- Bian, H., Li, C., Lin, Z., **Zhu, J.**, Chen, W., Li, G., Huang, Y., and Qu, Z. **Bacterial Turbulence in Shear Thinning Fluid.** *Phys. Rev. Research*, 8, 013318. DOI: <https://doi.org/10.1103/z22f-3v22>.
- Li, C., Bian, H., Qiao, Y., **Zhu, J.**, and Qu, Z. **Hydrodynamic Interaction Leads to the Accumulation of *Chlamydomonas reinhardtii* Near a Solid-Liquid Interface.** *Physical Review E*, 111, L052401. (Editor’s Suggestion) <https://doi.org/10.1103/PhysRevE.111.L052401>
- Zhu, J.**, Qiao, Y. (co-first author), Yan, L., Zeng, Y., Wu, Y., Bian, H., Huang, Y., Ye, Y., Huang, Y., Hii, C. W. R., Teng, Y., Guo, Y., Li, G., and Qu, Z. **Propulsion Contribution from Individual Filament in a Flagellar Bundle.** *Applied Physics Letters*, 126(7), 073702. 17 February 2025. <https://doi.org/10.1063/5.0243416>
- Cai, R., Niazi, M. D. K., Ai, Y., **Zhu, J.**, Wu, L., and Lu, X. **Fully Integrated Self-Propelling Microrobot in 180 nm CMOS with Sub-GHz Parity-Time-Symmetry On-Chip Energy Harvesting and Traveling Wave Electroosmosis Actuation.** *IEEE International Solid-State Circuits Conference (ISSCC)*, San Francisco, CA, USA, 2025, pp. 1–3. doi: [10.1109/ISSCC49661.2025.10904559](https://doi.org/10.1109/ISSCC49661.2025.10904559)

ABSTRACTS AND PRESENTATIONS

- Division of Fluid Dynamics Annual Meeting, Huston, TX.** 2025
Poster title: Propulsion Contribution from Individual Filament in a Flagellar Bundle.
- Division of Fluid Dynamics Annual Meeting, Huston, TX.** 2025
Presentation title: Complex Porous Surface Modulate Bacterial Turbulence.
- Chinese Congress of Theoretical and Applied Mechanics (CCTAM), Changsha (China)** 2025
Presentation title: Complex Porous Surface Modulate Bacterial Turbulence.
- The 13th Chinese Fluid Mechanics Academic Conference, Harbin (China).** 2024
Presentation title: Study on Microbial Flagellar Bundling Process and Propulsive Force Generation Based on Macroscopic Experimental Model.

SKILLS

Programming: MATLAB, C, C++, Python

Softwares & Tools: SolidWorks, LaTeX, COMSOL, Origin, LabVIEW, Adobe Illustrator

Prototyping: Arduino, 3D Printing (PLA, TPU, Resin), Laser Cutting

Biorelated skills: Bacteria culturing

WORK EXPERIENCE

Research Assistant, Yunker Lab, Georgia Institute of Technology *May 2026 - present*

Teaching Assistant, School of Physics, Georgia Institute of Technology *Aug 2025 - present*

- *PHYS-2212-F25, Principles of Physics II*

- *PHYS-2212-S26, Principles of Physics II*

Research Assistant, Bio-inspired Fluid Mechanics lab, Shanghai Jiao Tong University *Jan 2022 - Jul 2025*

Teaching Assistant, UM-SJTU Joint Institute, Shanghai Jiao Tong University *2021 - 2024*

- *VM320, Fluid Mechanics*

- *VM312, Mechanical Behaviour of Biological Tissues*

- *VM250, Design and Manufacturing I*

- *VG100, Introduction to Engineering*

AWARD & APPOINTMENT

- Co-founder and co-leader of ASME branch in UM-SJTU Joint Institute *Sep 2022 - Sep 2024*

- Outstanding Graduates of Shanghai (top 3% out of 350) *2023*

- ChunTsung Scholar (top undergraduate research program in SJTU) *2023*

- Capstone Project Gold Award at the UM-SJTU JI Design Expo (top 5% out of 50) *2022*

- The John Wu & Jane Sun Merit Scholarship (top 3% out of 350) *2022*

- Shanghai Jiao Tong University B-level Scholarship (top 5% out of 350) *2022*

- First Prize of Shanghai Mechanical Engineering Innovation Competition *2021*